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Effect of texture and microstructure on properties of electrodeposited Cu-SiO_2 and Cu-Y_2O_3 coatings

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ABSTRACT

In the present study, crystallographic texture evolution of copper, Cu-SiO_2 and Cu-Y_2O_3 coatings obtained by direct current (DC) and pulse direct current (PC) electrodeposition was correlated to their hardness and electrical conductivity. 30 g/l SiO_2/Y_2O_3 added acidic $CuSO_4$ solution was used as electrolyte and both DC and PC mode (5 and 10 kHz frequency, 30% duty cycle) depositions were carried out for 60 min). Coated specimens were characterized by X-ray diffraction for phase analysis and texture study. The effect of pulse parameter and second phase particles (SiO_2/Y_2O_3) on the crystallographic textures of [111], [001], and [110] orientations were analyzed. Coating morphology was studied with field emission scanning electron microscop (FESEM) whereas, surface-mechanical properties were investigated using microhradness testing. Electrical conductivity was also studied with four probe technique. Among the three deposition conditions (DC, 5 and 10 kHz), at 10 kHz pulsed frequency, the growth of [111] texture was observed in all systems. Highest hardness value of 240 HV, which is far better than bulk copper, was achieved for Cu-Y_2O_3 coating developed by 10 kHz pulse frequency. In spite of development of [111] fibre texture, minor decrease in electrical conductivity with increase in pulse frequency was observed in both the composite coatings and the same was attributed to finer matrix formation and inclusion of second phase insulating particles (SiO_2 and Y_2O_3) into the matrix.

1. Introduction

Due to excellent mechanical and corrosion resistance properties, electrodeposited composite coatings have attracted many researchers in the past decades [1–3]. Different types of composite and alloy coatings are manufactured by various deposition methods such as thermal plasma spray, physical vapor deposition (PVD), chemical vapor deposition (CVD) and electrodeposition [4]. In metal matrix composite coatings, different concentrations of the second phase reinforcement (oxides, carbides, metal particles, etc.) are used to get specific property. In case of direct current (DC) electrodeposition method, the coatings can easily be produced by changing the process parameters like the current density, stirring rate and particle concentrations in the bath [5–7]. Pulse current (PC) electrodeposition can produce diversified coating microstructure by varying the process parameters such as current density, pulse frequency ($=1/(T_{on} + T_{off})$) and duty cycle ($=T_{on}/(T_{on} + T_{off})$). It is one of the promising and economical techniques to prepare dense nanostructured coatings, in which the grain boundaries, grain size, and matrix texture can be precisely manipulated to achieve better mechanical and electrical properties [8].

High performance material that exhibits outstanding electrical conductivity along with high mechanical properties (hardness and wear resistance) is a requirement of electrical contacts as well as high density circuit boards used in microelectronic applications. Till today, Copper (Cu) is the most common material which is used for these applications and replacement by other is not completely successful [9]. Various efforts to strengthen copper by several methods including bulk alloying, nanocomposites formation, and deformation have been made [10–12]. But, these techniques led to a noticeable decrease in electrical conductivity of the material. Lately, nanoscale growths of electrodeposited and sputter-deposited Cu have been reported to show high hardness as well as good electrical conductivity [13]. For electrical applications, a highly (111) textured copper foil reveals better performance due to better hardness, electro-migration resistance, oxidation resistance, and lower internal stress compared to those with other textured Cu deposits [14,15].

Lu et al. reported copper foils of high electrical conductivity with a preferred (101) texture containing a high density of coherent twin boundaries, exhibiting a hardness of 2.0–2.6 GPa [16,17]. It was also documented that highly textured and twinned copper foils was prepared by PC deposition with a maximum hardness of 1.56 GPa and the reason behind lower hardness was reported as the void formation due to hydrogen entrapment at higher applied current densities [18]. Some reports based on (111) oriented electrodeposited Cu films have

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Table 1
Plating bath composition and deposition parameters.

Electrolyte (acidic copper sulfate bath)	Copper sulfate ($CuSO_4.5H_2O$): 200 g/l Sulfuric acid (H_2SO_4): 50 g/l
pH	2.17($\pm$0.02)
Current density	8A/dm^2
Pulse parameter/condition	No pulse (DC) and 5 and 10 kHz with 30% duty cycle
Temperature	Room temperature
Plating time	60 min
Dispersion	SiO_2/Y_2O_3: 30 g/l

also been published. Few years back, fabrication of (111) oriented and nano-twinned Cu by DC electrodeposition from chloride bath at high current density with enhanced mechanical properties was reported. They also reported 40 ppm chloride concentration as critical concentration at serious agitation, above which, (111) orientation was dominated [19]. But, Chen et al. reported about formation of (110) texture from same chloride bath with DC and PC deposition [20]. They have reported critical chloride concentration was about 40 ppm for DC electrodeposition without serious agitation and 60 ppm for pulse electrodeposition. But the effect of orientation on mechanical and electrical properties of composite coatings was not explored systematically.

The main objective of this study is to investigate the microstructure and crystallographic texture of pure Cu, Cu-SiO_2 and Cu-Y_2O_3 coatings produced by direct and pulse current electrodeposition for better understanding of their microstructural, surface-mechanical and electrical characteristics and correlations between them. This is the first reported trial, where orientation dependent surface-mechanical and electrical properties of the above said composite coatings have been thoroughly studied. The aim of such a detailed crystallographic characterization was to determine the effect of DC and pulse parameters as well as second phase oxide nano particles (SiO_2 and Y_2O_3) on orientation development and their influence on surface-mechanical and electrical behavior of the developed coatings.

2. Experimental procedure

Pure Cu, Cu–SiO_2 and Cu–Y_2O_3 coatings were electrodeposited on copper substrates of specific dimensions (20 mm × 10 mm × 2 mm). SiO_2 powder (Sigma Aldrich, China) of size 15–20 nm and Y_2O_3 power (Alfa Aesar, USA) of size 10–15 nm were procured and used directly in the deposition process without further processing. The chemical compositions of the electrolyte and deposition parameters are presented in Table 1. Copper plates were used as both anode and cathode (substrates). The substrates were metallographically polished and cleaned with acetone followed by thorough wash with distilled water prior to the deposition process. Anode was placed parallel with cathode (substrate) in the plating bath. Each deposition process was conducted at room temperature for 60 min. The bath was constantly agitated with magnetic stirrer during the deposition process for complete dispersion of second phase particles in the electrolyte. The electrolyte was also sonicated for 15 min before deposition process for the same purpose.

The phase analysis of the coatings was carried out by X-ray diffractometer (XRD) (Rigaku Ultima-IV) with Cu *Kα* radiations. XRD patterns were recorded under ambient conditions at a scan speed of 10° per min. The macrotexture of all the deposited specimens was measured in a fully automatic X-ray texture goniometer fitted in a Bruker D8 Discover Davinci System using Cu *Kα* radiation. Three pole figures ((111), (200), and (220)) were measured and the orientation distribution function (ODF) was calculated by using Labotex 3.0 software [21]. The volume

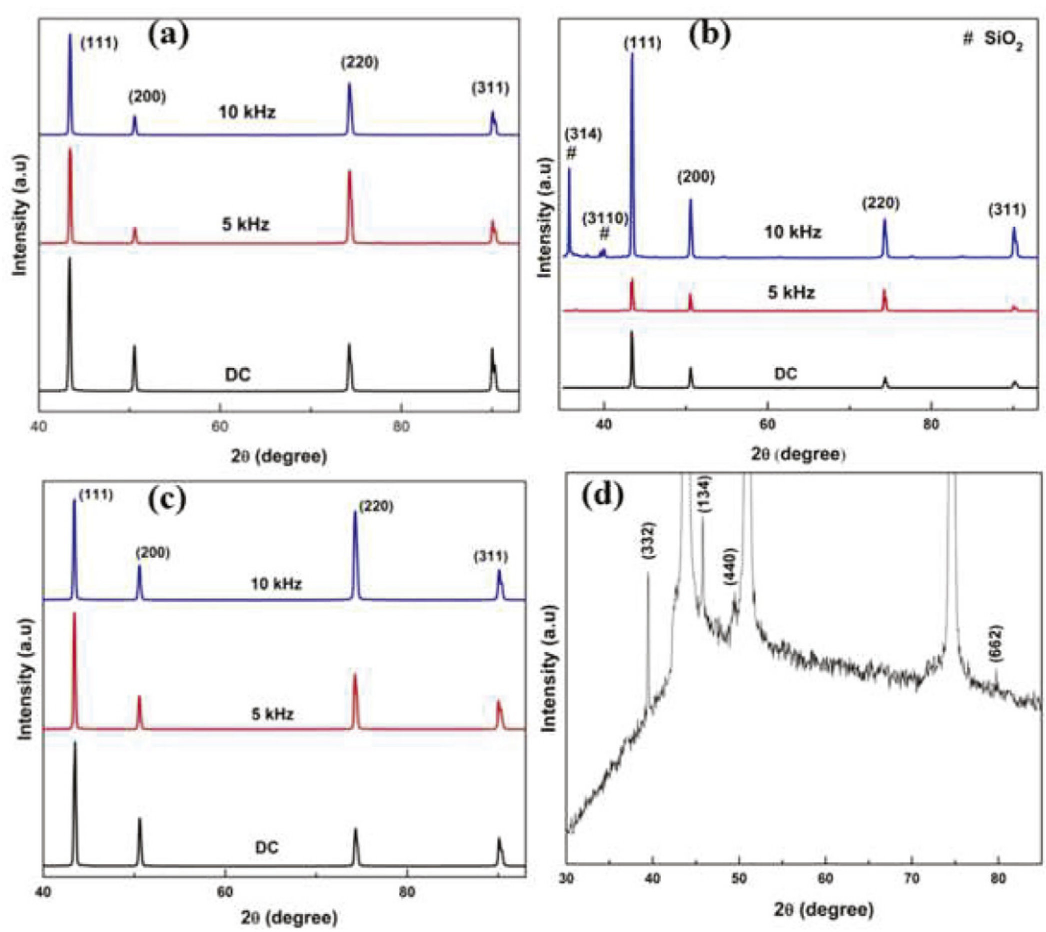

Fig. 1. XRD pattern of (a) pure copper (b) Cu-30 g/lSiO$_2$ and (c) Cu-30 g/lY$_2$O$_3$ coatings (d) enlarge view of Cu-Y$_2$O$_3$ composite coating (10 kHz frequency).

Table 2
Lattice strain (%) of developed coatings.

Samples	Planes			
	(111)	(200)	(220)	(311)
Pure Cu (DC)	0.484	0.562	0.506	0.505
Pure Cu (5 kHz)	1.650	1.630	1.014	0.855
Pure Cu (10 kHz)	1.278	1.195	0.811	0.685
Cu-SiO_2 (DC)	0.576	0.614	0.494	0.481
Cu-SiO_2 (5 kHz)	0.913	0.844	0.613	0.556
Cu-SiO_2 (10 kHz)	1.110	1.073	0.796	0.686
Cu-Y_2O_3 (DC)	1.205	1.218	0.863	0.743
Cu-Y_2O_3 (5 kHz)	1.435	1.307	0.912	0.735
Cu-Y_2O_3 (10 kHz)	1.862	1.632	1.044	0.843

fractions of different fibre orientations were also estimated through integration method with 5° tolerance on the deviation from the ideal orientations [21]. The surface morphology of the coatings was determined with the help of Nova Nano-450 model field emission scanning electron microscope (FE-SEM). Obtained micrographs were qualitatively analyzed by Axio Vision Release 4.8 service pack-2 image analysis software to measure the grain size distribution and average grain size of the developed coatings. The microhardness of pure Cu, Cu–SiO_2 and Cu–Y_2O_3 coatings was determined using a Vickers's microhardness tester (LECO LM700) with 10 g load. Each hardness value reported in the current work is an average of minimum 5 measurements obtained at equivalent locations on the surface of the coated specimens. High precision KEITHLEY 6221 current source combined with 2182A nano-voltmeter in a four probe setup was used for measurement of electrical conductivity of the deposited specimens.

3. Results and discussion

3.1. XRD analysis

Fig. 1(a–c) show the XRD spectra of the prepared pure copper, Cu-SiO_2 and Cu-Y_2O_3 coatings under various deposition conditions (DC, Pulsing with 5 and 10 kHz) respectively. All Cu peaks were assigned by JCPDS (card no. 04–0836) data bank. Some peaks of SiO_2 were observed in Cu-SiO_2 composite coating due to the presence of silica mainly at 5 and 10 kHz pulse frequency. In case of Cu-Y_2O_3 composite coating no Y_2O_3 peak was observed normally, but by enlarging the XRD profile of one such composite coating, small Y_2O_3 peaks were observed.

Lattice strain of the developed coatings was calculated by the following formula [22] and the values were tabulated in Table 2.

$$\delta\% = \frac{d_{substrate} - d_{coating}}{d_{substrate}} \times 100 \quad (1)$$

where $d_{substrate}$ = d-spacing of the particular plane in the substrate (Cu), $d_{coating}$ = d-spacing of the same plane of the coating (Cu matrix of the coating).

From the table it was observed that in case of composite coatings, lattice strain was increased in all planes with change in deposition conditions from DC to 5 kHz pulse frequency and further increased for 10 kHz pulsing. But, in pure copper coating, strain is highest at 5 kHz pulse frequency followed by 10 kHz and DC deposition. In general, it was also observed that lattice strain is more in case of Cu-Y_2O_3 composite coatings.

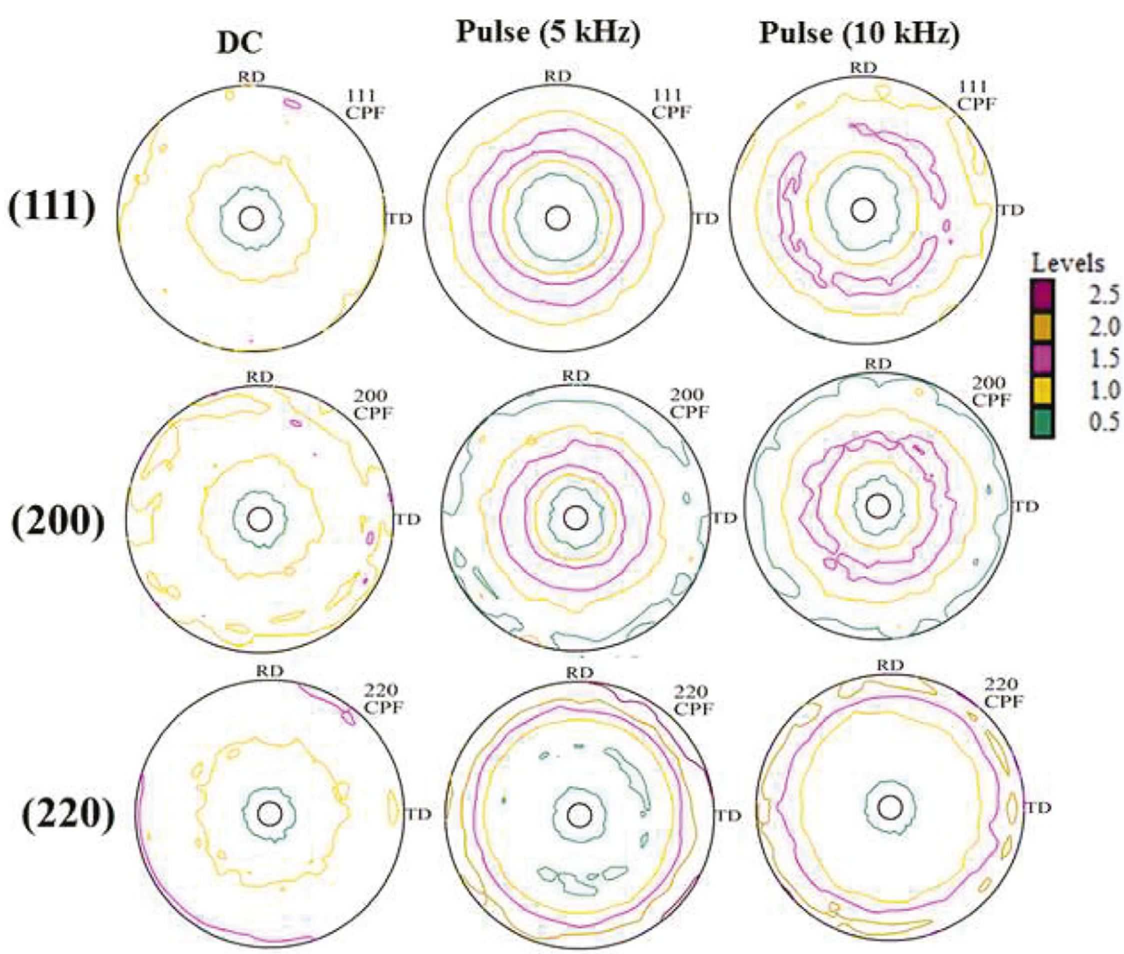

Fig. 2. (111), (200) and (220) pole figures of pure copper coatings with DC and pulse variation of 5 and 10 kHz.

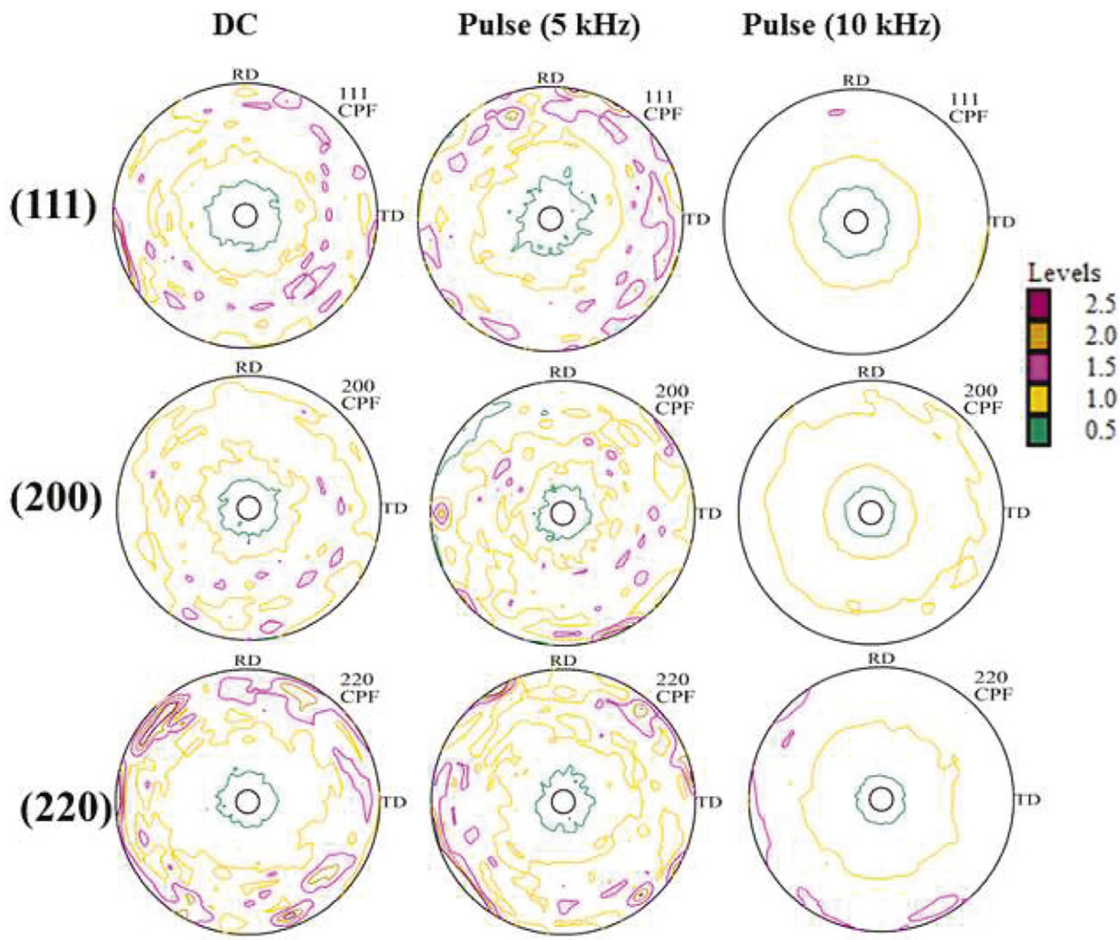

Fig. 3. (111), (200) and (220) pole figures of Cu-30 g/l SiO_2 coatings with DC and pulse variation of 5 and 10 kHz.

3.2. Texture study

Fig. 2 shows the texture development of electrodeposited copper coatings in terms of (111), (200) and (220) pole figures as a function of deposition conditions such as DC, 5 and 10 kHz pulse frequency. It can clearly be seen that, in case of DC deposition, [111] fibre texture was developed. When the electrodeposition condition was changed to PC, the initial intensities of [001], [110] and [111] orientations were decreased with increase in pulse frequency; but, the [111] orientation remained as preferred orientation. Earlier it was reported that, the texture development in thin and thick coatings is largely depending upon the effect of the substrate and the influence of the surface energy on the growth rate [23]. During the deposition process strain energy, surface energy or a combination of both surface and strain energy and the energy arising from various interfaces are considered as controlling factors in the crystallographic texture formation of FCC metals [24]. In addition to this, the crystallographic texture of the coatings is also influenced by various process parameters such as *pH* of the electrolyte, surfactants, pulse frequency, applied current densities, thickness of the deposits, and formation of stress/strain in the deposited specimens [25]. Change of crystallographic orientation from (111) to either (100) or (101) in most cases is subjected to the changes that happen during the deposition process [26,27]. Besides this, the stability of the surface plane of the metal also has a great influence on evolution of crystallographic orientations [28,29]. As the surface energy minimization in FCC metals follows the order $\gamma(101) > \gamma(100) > \gamma(111)$, in general the stability of the individual surface plane during texture formation follows the order (111) > (100) > (101) [30,31]. Thus, from earlier literature it can be told that, due to minimization of energy, a dominant [111] fibre texture would be expected to develop during the electrodeposition process. This may be considered as the reason behind the development of [111] orientation in pure copper coating irrespective of deposition conditions or techniques used. The low surface energy of [111] orientation mainly driven by various factors such as the coordination number [32] and atomic density compared to (100) and (101) planes [33,34]. Moreover, low deposition temperatures and high current density are considered as other factors to minimize the surface and interfacial energies of the material, which favors growth of (111) orientation [35]. This can also be considered as partial explanation for formation of [111] orientation in pure Cu coating, as the process was carried out with relatively high current density (8 A/dm^2) at room temperature.

Figs. 3 and 4 displays the (111), (200) and (220) pole figures of Cu-SiO_2 and Cu-Y_2O_3 composite coating respectively. Fig. 3 describes mostly random texture during DC deposition. However, with the change in deposition condition from DC to 5 kHz pulse, the [111] fibre orientation was found to be increased whereas the other orientations ([001] and [110]) were decreased with the increase in pulse frequency. With further increase in pulse frequency to 10 kHz, [111] fibre orientation became more pronounced. Similar observations were also made in the case of Cu-Y_2O_3 composite coatings (Fig. 4).

Earlier it was also reported that, during deposition process, (100) orientation growth is marginally favored at larger off times [36] and (101) is known to be favored at very large off times [16]. This may occur by consuming the (111) grains, which is possible due the stress-driven recrystallization during pulse electrodeposition [37]. Moreover, during PC deposition with longer off time, formation of (101) texture is highly favorable [38]. But, though in the present study, depositions were carried out at high pulse frequency (5 kHz) or lower off time

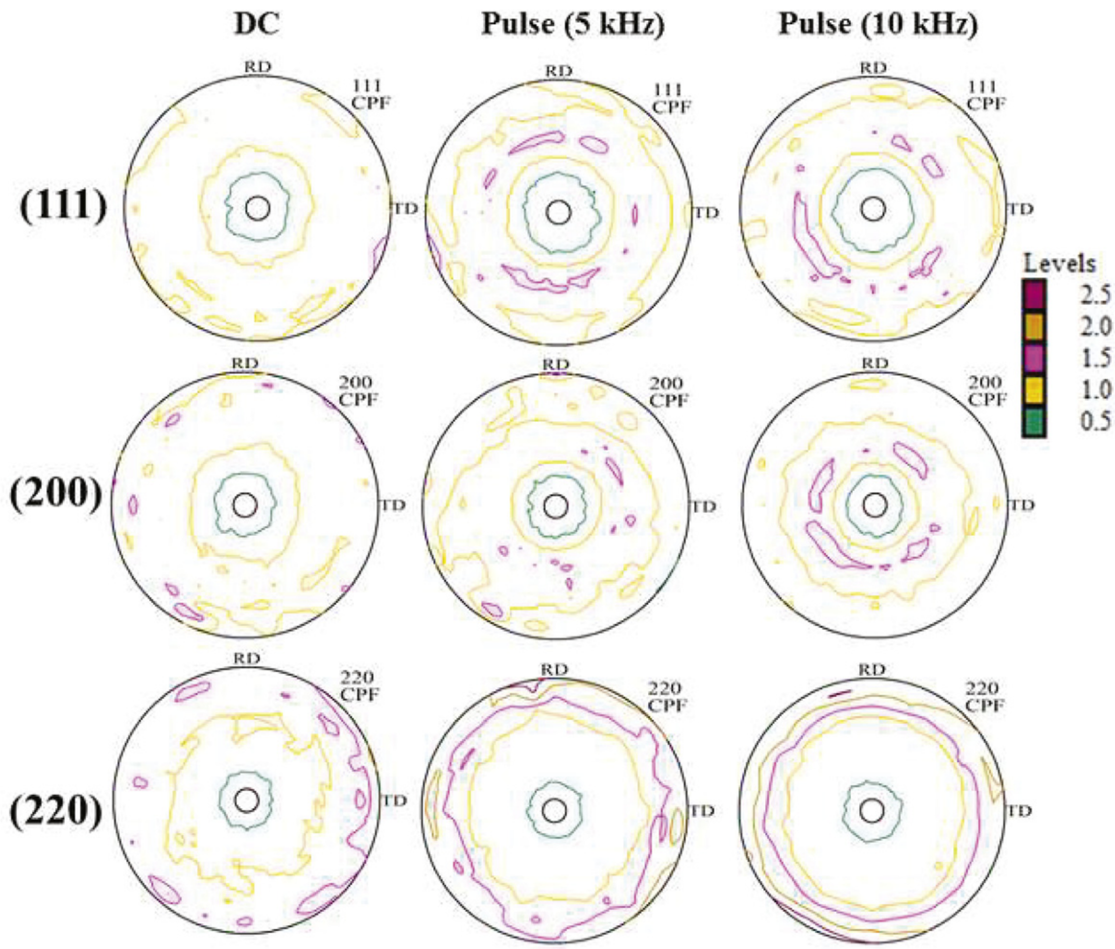

Fig. 4. (111), (200) and (220) pole figures of Cu-30 g/l Y_2O_3 coatings with DC and pulse variation of 5 and 10 kHz.

and further lower off time in case of 10 kHz, [111] orientation found prominent in both Cu-SiO_2 and Cu-Y_2O_3 composite coatings. High strain energy might be the cause of favoring such orientation during these high frequency depositions.

To quantify the texture development in the deposited specimens, ODF figures of all the coatings are shown in Fig. 5. The ODF figures are considered at $\phi_1 = 0°$ and on ($\phi_2 - \phi$) plane. From the figure, development of [111] orientation along with [001] and [110] was observed in pure copper DC mode deposition. The preferred [111] orientation was centered at $\phi_1 = 0°$, $\phi_2 = 45°$ and $\phi = 55°$, whereas other two orientations ([001] and [110]) were centered at $\phi_1 = 0°$, $\phi_2 = 3°$ and $\phi = 5°$ and $\phi_1 = 0°$, $\phi_2 = 45°$ and $\phi = 0°$ respectively. Further, with increased pulse frequency, the [001] and [110] orientations became absent and existence of only [111] orientation was observed. The higher intensity [111] fibre orientation was regarded as the final and main texture component in pure copper coated samples. From the ODF figure it was also observed that at DC deposition condition, the orientation distribution is more of random type in Cu-SiO_2 and Cu-Y_2O_3 composite coatings with [001] having marginal higher intensity. But after changing the deposition conditions from DC to 5 kHz pulse frequency, the development of [111] fibre orientation was observed in all composite coated specimens and further intensification of [111] fibre orientation was also witnessed at 10 kHz by consuming other [001] and [110] orientations.

To further quantify the texture development, volume fraction of the [111], [001] and [110] orientations was measured and reported in Fig. 6. Fig. 6a shows the volume fraction of pure copper coatings. The obtained result shows higher volume fraction of [111] orientation at DC condition, which was confirmed as strong [111] fibre. But, with the change in deposition condition from DC to pulse deposition of 5 and 10 kHz, the volume fraction of [001] and [110] orientation was decreased rapidly as compared to [111] fibre orientation. Though the volume fraction of the [111] fibre is marginally decreased, it is the only orientation which exists after the pulse variation. Fig. 6b shows the volume fraction of Cu-SiO_2 composite coating. The figure clearly indicates random texture for DC deposition with [001] orientation having marginal higher volume fraction. This orientation was attributed may be due to the change in surface energy of the fibre orientations for addition of second phase nano particles. Moreover, addition of second phase particles induces higher strain energy in case of both composite coatings compared to pure copper coating, which can be confirmed from the Table 2. This can favour the formation of [001] and [110] orientations [39,40]. But, when the process parameter was changed from DC to PC mode, the volume fraction of [001] and [110] was decreased and at the same time the volume fraction of [111] orientation was increased, which signifies that [111] fibre was the sole preferable orientation. Similarly, Fig. 6c illustrates the change in volume fraction of [111], [001] and [110] fibre orientation with respect to the different process parameter in Cu-Y_2O_3 composite coating. The trends of different fibre textures were exactly similar to that of Cu-SiO_2 coatings. But the volume fraction of [111] fibre was higher in Cu-Y_2O_3 than Cu-SiO_2 specimen due to higher strain.

3.3. Microstructural characterization

The FESEM micrographs of all deposited specimen are presented in Fig. 7(a–i). The obtained morphology of different samples gives comprehensive information about the grain structure with respect to deposition parameters and second phase oxide (SiO_2 and Y_2O_3) nanoparticle incorporation. Fig. 7(a–c) shows the morphology of pure copper coating

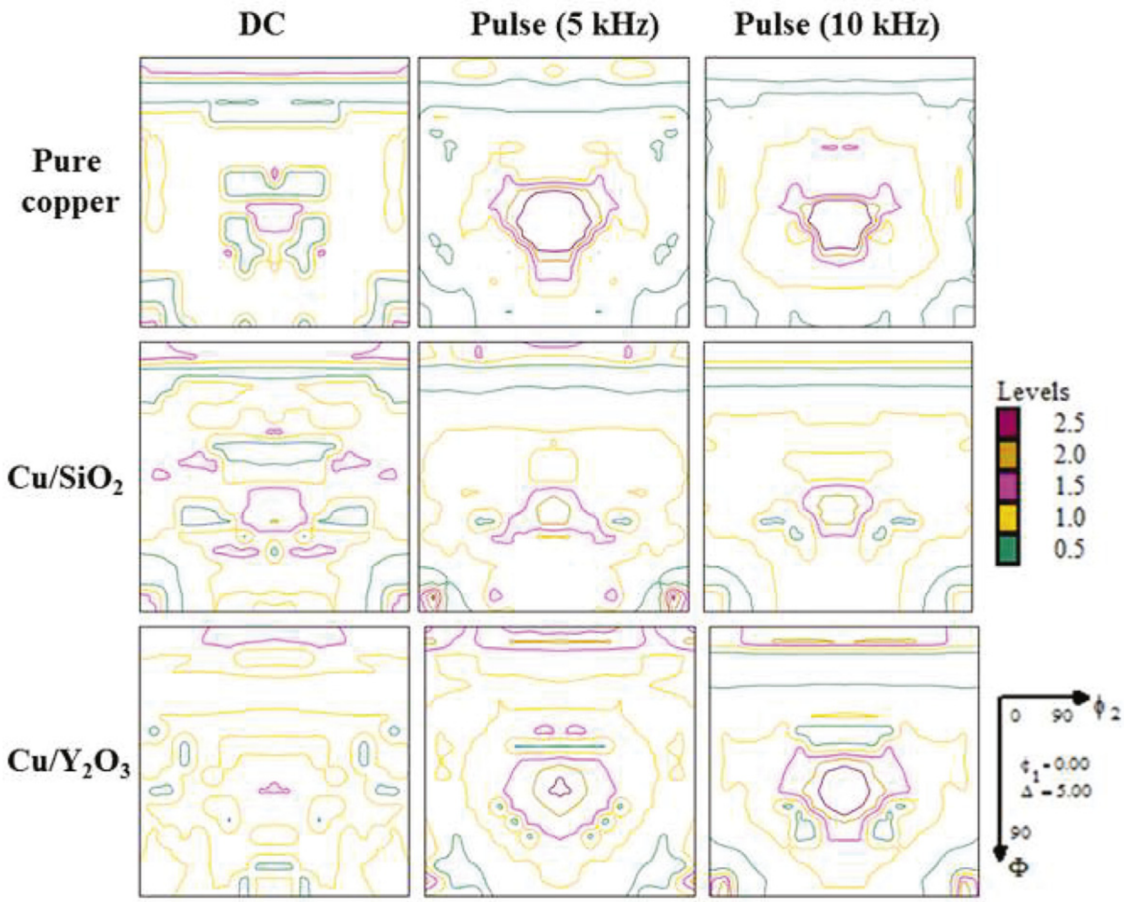

Fig. 5. ϕ_1 = constant ODF figure of pure copper, Cu-30 g/l SiO_2 and Cu-30 g/l Y_2O_3 composite coatings with DC, pulse deposition of 5 and 10 kHz.

at different deposition parameters (DC, 5 and 10 kHz) respectively. In case of different pulsing rates (Fig. 7b and c), reasonably finer and more uniform structures than that of direct current deposition was observed. Pure copper coating obtained with DC deposition showed ~10 μm of average grain size as reported in Table 3. Decreasing trend of average grain size with increasing pulse frequency was observed and was considered as obvious due to better nucleation mechanism. Higher pulse frequency means the pulses are shorter, i.e., both T_{on} and T_{off} are of short duration. This specifies that double layers do not get enough time to be fully charged and discharged during T_{on} and T_{off} time respectively. Due to shorter pulses, the pulse arrives very quickly as soon as the previous pulse cycle is completed. Due to this mechanism, thin pulse diffusion layer is produced, which obstruct the transportation and diffusion of the migrating copper ions from the electrolyte to the surface of the copper substrate. This leads to the superior nucleation with limited growth rate resulting finer matrix formation in all the coatings prepared with higher pulse frequency [41].

After addition of 30 g/l SiO_2 into the deposition bath, the structure became coarser and the average grain size increases. This may be explained by increaase in surface energy of the Cu deposition due to adverse nucleation effect by addition of SiO_2 in the electrolyte. This leads to excess growth of Cu crystals as well as less nucleation which resulted coarse grain size than pure copper coating. The formation of Cu-Cu cluster is more expected compared to Cu-SiO_2 cluster as evident from the first-principle calculations and the same has been reported earlier for Cu-SiO_2 system [41].

But, in case of Cu-30 g/l Y_2O_3 system (Fig.7g–i), the structure looks more compact and finer than other coatings. The same can be validated from the value reported in Table 3. The uniform codeposition of Y_2O_3 particles hinders the grain growth in the Cu matrix resulting in smaller grain sizes, which is completely opposite to the Cu-SiO_2 coating. In order to produce compact and finer structure, controlled nucleation of the crystals should occur and this can be realized by various ways, for example: (a) adding appropriate grain refining agents to the deposition bath, (b) adding a second phase reinforcement and/or (c) pulse current deposition. Addition of second phase Y_2O_3 nano particles leads to change in the shape of copper crystallites to more compact and finer morphology and this in turn reduces their average grain size (Table 3).

Energy dispersive spectra of Cu-SiO_2 and Cu-Y_2O_3 composite coatings were displayed in Fig. 8a and b respectively. Fig. 8a confirmed the presence of Cu along with Si and O, whereas the presence of Cu, Y and O is clear from Fig. 8(b). Semi-quantitative investigation of the spectra was carried out for all the Cu-SiO_2 and Cu-Y_2O_3 deposited specimens. The amount of Si was converted to equivalent amount of SiO_2 using the stoichiometric ratio of Si and O as per chemical formula of SiO_2. Similar calculation was also used for conversion of amount of Y to Y_2O_3. Fig. 8(c) shows weight percentage of SiO_2 and Y_2O_3 in the deposited composite coatings at different deposition conditions. Each value reported in Fig. 8(c) was actually average of 5 EDS spectra. The amount of SiO_2 and Y_2O_3 wt.% in the composite coating increases with the increase in pulse frequency.

3.4. Microhardness and electrical conductivity study

Fig. 9(a) shows the hardness of copper, Cu-30 g/l SiO_2 and Cu-30 g/l Y_2O_3 coatings as a function of pulse frequency. The hardness values were found to be increasing by changing the deposition parameter from DC to 5 kHz and further changing to 10 kHz pulse frequency. Among all deposited specimens, the best hardness result was observed at 10 kHz pulse frequency. In case of electrodeposited Cu and Cu-based

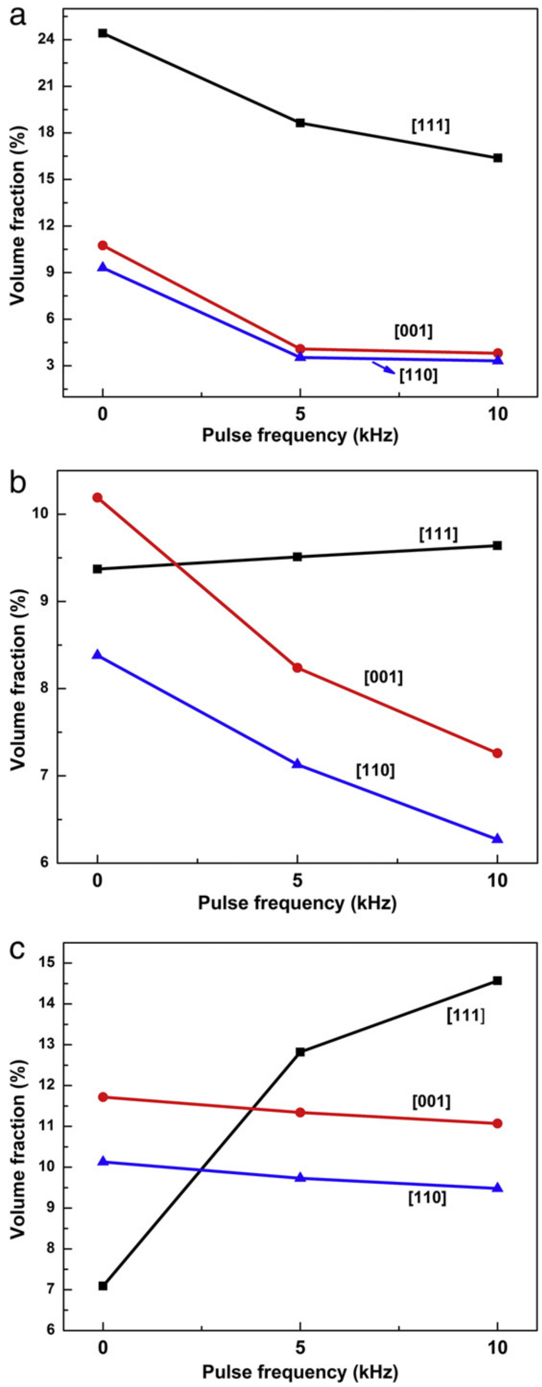

Fig. 6. Volume fraction of different orientations in (a) pure copper (b) Cu-30 g/l SiO_2 and (c) Cu-30 g/l Y_2O_3 coatings.

composite coatings, improvement of hardness can be attributed to various reasons such as (i) dispersion strengthening due to addition of second phase nano particles (SiO_2 and Y_2O_3), (ii) formation of finer matrix caused by nucleation mechanism due to altering the deposition parameters (pulse frequency, current density etc), (iii) development of specific crystallographic orientation with increase in pulse frequency, i.e. [111] fibre orientation and (iv) lattice strain as observed in Table 2. Development of [111] orientation is generally considered as harder orientation compared to other textures because when a load is applied on the (111) plane, it is less easy to activate the $\{111\}\langle 01\bar{1}\rangle$ slip systems of the material compared to other orientations. Though, no second phase particle was added in the bath during the deposition of pure copper coating, the improvement in hardness at 10 kHz compared to 5 kHz and DC deposition can only be explained by the latter two mechanisms. But in case of both Cu-SiO_2 and Cu-Y_2O_3 composite coatings, the increased hardness with increasing pulse frequency can be attributed to all three mechanisms, i.e. increased amount of co-deposited SiO_2 and Y_2O_3 causing dispersion hardening, formation of finer matrix due to better nucleation and existence of [111] preferred orientation at higher pulse frequency. Moreover, it can also be observed that in general with increasing lattice strain of the composite coatings, the hardness increases. Among all the coatings, Cu-Y_2O_3 composite coating displayed better hardness compared to pure Cu and Cu-SiO_2 deposits. The higher hardness of Cu-Y_2O_3 coating can be explained by various factors such as; (i) higher amount of codeposited Y_2O_3 at all the deposition conditions than Cu-SiO_2 coatings, (ii) lower average grain size in all cases than both pure Cu and Cu-SiO_2 coating, (iii) higher volume percentage of [111] orientation in case of Cu-Y_2O_3 composite coating compared to Cu-SiO_2 coating.

Electrical conductivity (in terms of The International Annealed Copper Standard or IACS) of copper coatings, Cu-30 g/l SiO_2 and Cu-30 g/l Y_2O_3 composite coatings are displayed in Fig. 9(b). In case of pure Cu coating, very minor decrease in electrical conductivity was observed with changing the deposition condition from DC to pulse. It was previously well documented that in spite of finer matrix, formation of [111] orientation favors increase in electrical conductivity of the material [42,43]. High electrical conductivity of (111) oriented coatings was reported due to the small angle grain boundaries, which exhibit one order lower electrical resistivity compared to that of high angle grain boundaries [12,16], and the highly uniform orientation of the grains that reduces the electron scattering. In addition, the absence of impurities due to the surfactant free sulfate bath might be contributed to the better electrical conductivity. This can be the reason behind the almost constant electrical conductivity values of pure copper coating in spite of decreasing grain size with increasing pulse frequency. But, this cannot be the reason for the case in Cu-SiO_2 and Cu-Y_2O_3 composite coatings, as in spite of increase of [111] preferred orientation in these coatings, noticeable decrease in electrical conductivity was observed. This can be explained by formation of finer matrix due to faster nucleation mechanism and addition of insulating oxides (SiO_2 and Y_2O_3) in the coatings. Comparing all the deposits, pure copper coating displayed best electrical conductivity followed by Cu-SiO_2 and Cu-Y_2O_3 composite coatings.

4. Conclusions

Pure copper, Cu-30 g/l SiO_2 and Cu-30 g/l Y_2O_3 composite coatings were developed by electro-codeposition process. The effect of texture and microstructure of the coatings and incorporated second phase oxide nanoparticles on the microhardness and electrical conductivity were studied systematically. The key inferences are:

- Better hardness was observed at 10 kHz pulse frequency followed by 5 kHz in all types of deposits. This was due to development of (111) texture and higher amount of codeposited second phase particles in the matrix as well as formation of finer matrix at higher pulse frequency.
- Cu-Y_2O_3 composite coating displayed best hardness result compared to other deposits due to higher volume fraction of [111] orientation, finer grain size as well as more amount of codeposited Y_2O_3 particle than Cu-SiO_2 composite deposits.
- Formation of [111] preferred crystallographic orientation of the coating matrix was observed for all deposited specimens. This affects the surface mechanical and electrical properties of the prepared coating.
- Though the existence of preferred [111] orientation favors better electrical conductivity of coated samples, decrease in electrical conductivity with increasing pulse frequency of composite coated specimens was observed due to formation of finer matrix as well as addition of second phase insulating particles.

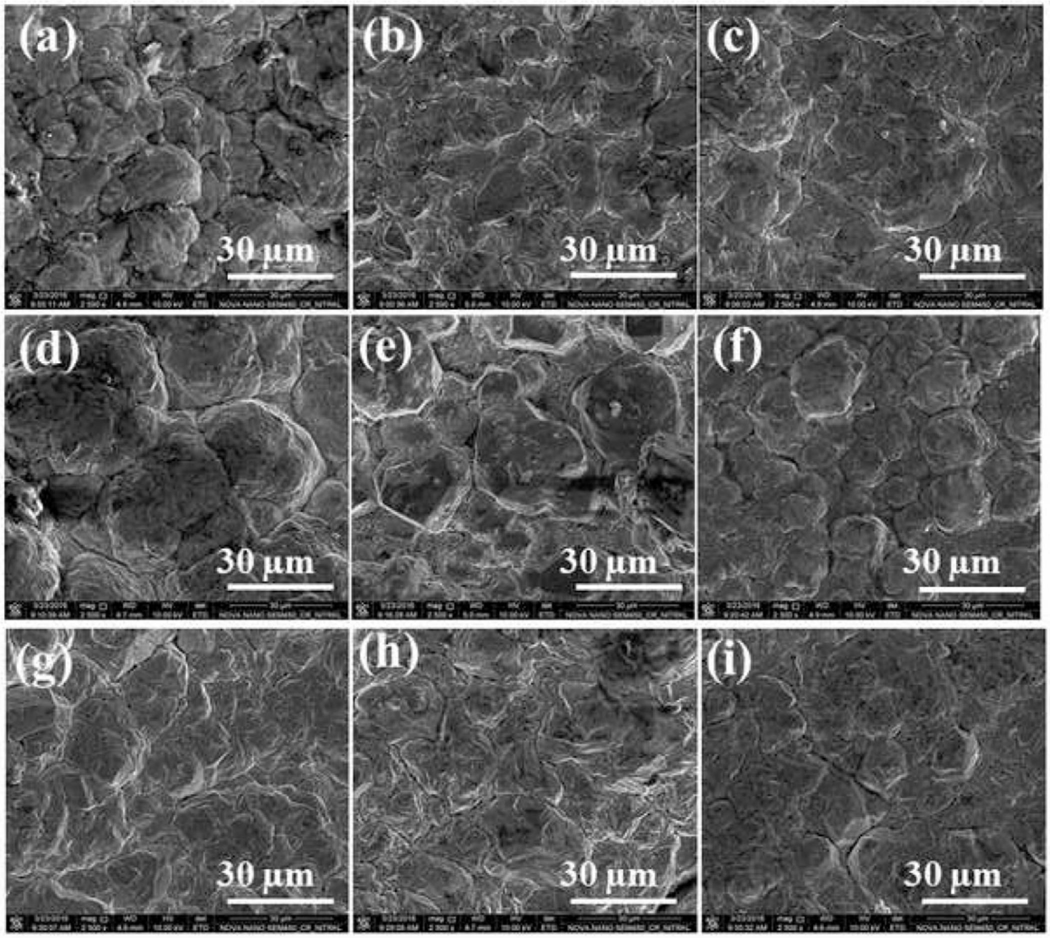

Fig. 7. SEM images of (a, b, and c) pure copper, (d, e, and f) Cu-30 g/l SiO_2 and (g, h, and i) Cu-30 g/l Y_2O_3 coatings with DC, 5 kHz and 10 kHz respectively.

Table 3
Average grain sizes (in μm) of all the deposits.

Deposited specimens	DC deposition	Pulse deposition (5 kHz)	Pulse deposition (10 kHz)
Pure copper	10.49	9.95	8.34
Cu-30 g/l SiO_2	13.30	12.36	11.19
Cu-30 g/l Y_2O_3	10.17	8.84	7.90

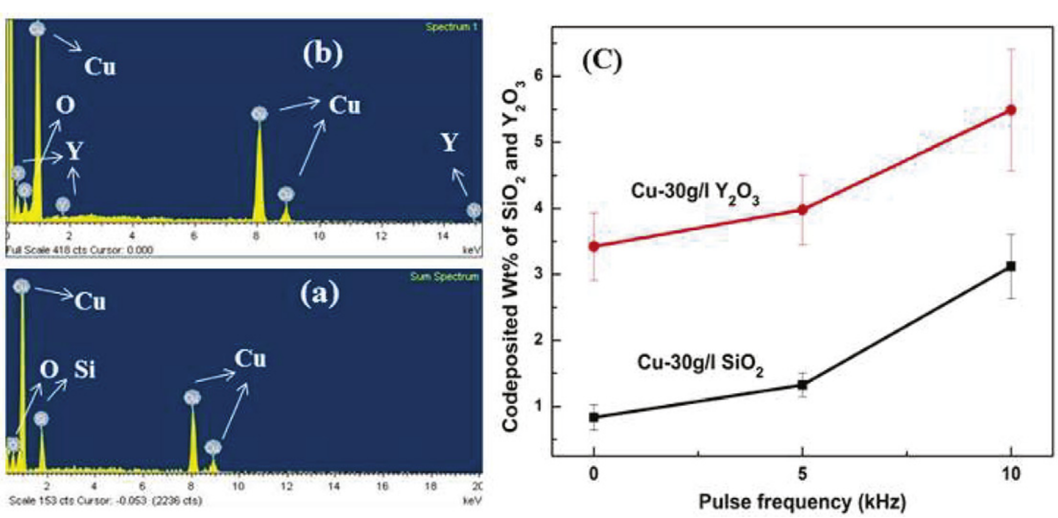

Fig. 8. (a and b) EDS spectra of Cu-SiO_2 and Cu-Y_2O_3 at 10 kHz pulse frequency respectively, (c) Co-deposited Wt.% of SiO_2 and Y_2O_3 verses pulse parameter.

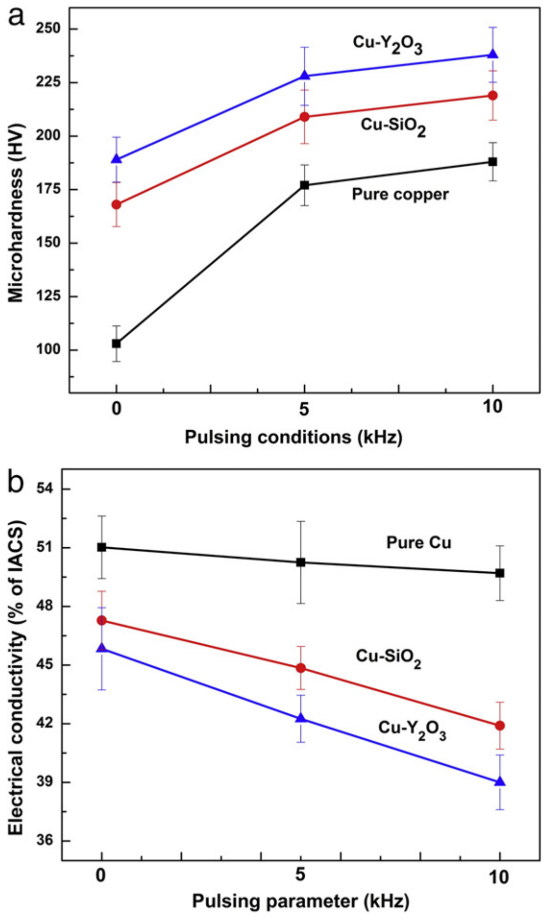

Fig. 9. (a) Microhardness and (b) Electrical conductivity of all deposited specimens.

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